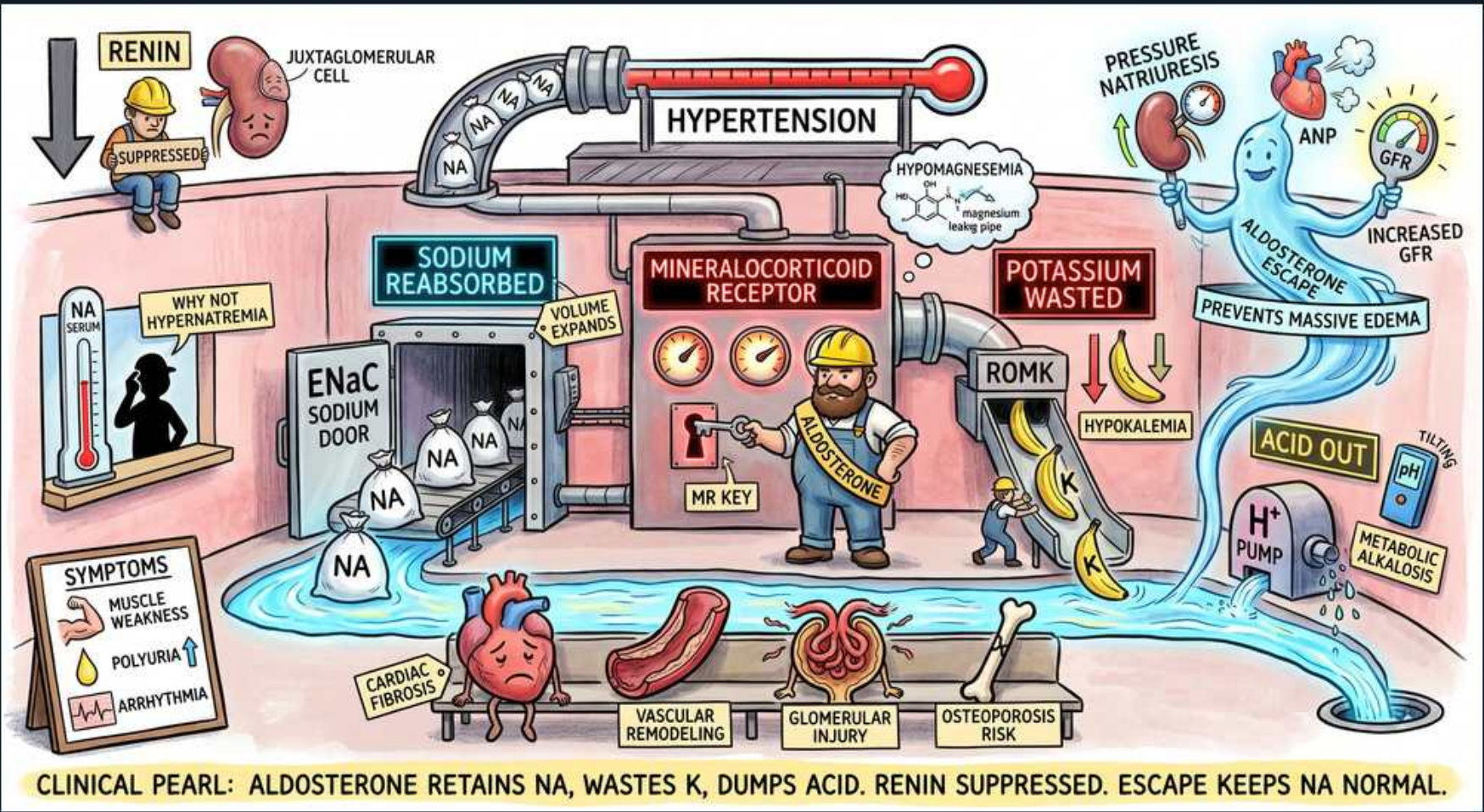


Primary Aldosteronism — The Aldosterone Salt Factory



1. Suppressed renin

WHAT YOU SEE

A glum, downturned kidney (the juxtaglomerular cell) labeled SUPPRESSED under a big down-arrow for RENIN — the JG cell shut off by the volume expansion is renin being suppressed, the fingerprint of PRIMARY aldosteronism.

WHAT IT ENCODES

Aldosterone is normally driven by angiotensin II and by hyperkalemia acting on the zona glomerulosa. In PRIMARY aldosteronism the adrenal secretes aldosterone autonomously, so the resulting Na/volume expansion feeds back to the juxtaglomerular cells and SUPPRESSES renin. The hallmark biochemistry is therefore HIGH aldosterone with LOW (suppressed) renin → an elevated aldosterone:renin ratio (ARR). This suppressed renin is exactly what separates primary from secondary aldosteronism.

BOARD TRAP

Renin is HIGH — that describes SECONDARY aldosteronism (renal artery stenosis, renin-secreting tumor, diuretics, heart failure/cirrhosis), where high renin drives high aldo. Primary aldo = LOW renin.

2. Hypertension pipe

WHAT YOU SEE

Big red HYPERTENSION pipe across the top

WHAT IT ENCODES

Na and volume retention drive the hypertension, which is the dominant clinical feature and is often RESISTANT (uncontrolled on ≥3 drugs including a diuretic). Primary aldosteronism is the most common SECONDARY/curable cause of HTN (5-10% of hypertensives, up to ~20% of resistant HTN). Aldosterone also causes direct vascular/cardiac damage beyond the BP effect.

BOARD TRAP

Dismissing it as essential HTN — young-onset HTN, resistant HTN, or HTN with hypokalemia/adrenal incidentaloma all mandate ARR screening.

3. ENaC sodium door

WHAT YOU SEE

ENaC labeled door letting NA sandbags through

WHAT IT ENCODES

Aldosterone binds the cytoplasmic mineralocorticoid receptor in principal cells of the cortical collecting duct; the activated receptor increases transcription of the epithelial sodium channel (ENaC) on the luminal membrane and the Na-K-ATPase on the basolateral side. Na is reabsorbed through ENaC. ENaC is the channel blocked by amiloride/triamterene and constitutively activated in Liddle syndrome.

4. Sodium reabsorbed

WHAT YOU SEE

Blue SODIUM REABSORBED sign, NA bags

WHAT IT ENCODES

Na reabsorption through ENaC expands extracellular volume and raises blood pressure. Reabsorbing Na also creates a lumen-negative electrical gradient that DRIVES the secretion of K⁺ (via ROMK) and H⁺ — this is why hypokalemia and metabolic alkalosis accompany the HTN.

5. Mineralocorticoid receptor

WHAT YOU SEE

Gauge panel labeled MINERALOCORTICOID RECEPTOR with Mr Key

WHAT IT ENCODES

The mineralocorticoid receptor (MR) is the molecular target of aldosterone and the therapeutic target of MR antagonists: spironolactone (non-selective, causes gynecomastia/menstrual irregularity) and eplerenone (selective, fewer anti-androgen effects). The MR cannot itself distinguish cortisol from aldosterone — protection comes from 11β-HSD2 converting cortisol to inactive cortisone (its failure causes AME).

6. Potassium wasted (ROMK)

WHAT YOU SEE

POTASSIUM WASTED chute, bananas down ROMK slide to HYPOKALEMIA

WHAT IT ENCODES

K⁺ is secreted into the tubular lumen through ROMK channels, driven by the lumen-negative voltage from Na reabsorption, producing renal potassium wasting and hypokalemia. Classic findings include K <3.5, inappropriately high urinary K (>30 mEq/day) despite hypokalemia.

BOARD TRAP

Requiring hypokalemia to diagnose — MOST patients with primary aldosteronism are NORMOKALEMIC. Normal K does NOT rule it out; screen with ARR regardless.

7. Acid out / alkalosis

WHAT YOU SEE

H⁺ PUMP pushing acid out, METABOLIC ALKALOSIS pH bottle

WHAT IT ENCODES

Aldosterone stimulates H⁺ secretion by the type A intercalated cell H⁺-ATPase, and the hypokalemia drives intracellular H⁺ shift plus increased renal ammoniagenesis — together producing a hypokalemic METABOLIC ALKALOSIS (high bicarbonate). Classic primary aldo labs: HTN + hypokalemia + metabolic alkalosis + low renin.

8. Why not hypernatremia

WHAT YOU SEE

Question over a thermometer of NA

WHAT IT ENCODES

Despite avid Na reabsorption, serum sodium stays in the normal-to-high-normal range rather than frankly elevated. This is because of aldosterone escape (below) and ADH-driven water retention that dilutes the Na. A mildly high-normal Na in a hypertensive patient is actually a subtle clue toward primary aldo.

BOARD TRAP

Expecting overt hypernatremia — it is the absence of dramatic hypernatremia/edema (due to escape) that is characteristic.

9. Aldosterone escape

WHAT YOU SEE

Smiling ANP figure, ALDOSTERONE ESCAPE, increased GFR

WHAT IT ENCODES

Aldosterone escape: after initial Na/volume gain, pressure natriuresis (raised BP increases renal Na excretion) plus atrial natriuretic peptide (ANP, released by atrial stretch) restore Na balance. This prevents progressive volume overload and massive edema — so primary aldosteronism causes HTN WITHOUT edema. Note escape applies to the Na/volume axis only; K wasting CONTINUES.

BOARD TRAP

Expecting peripheral edema — its absence (because of escape) is the rule, not the exception.

10. Hypomagnesemia

WHAT YOU SEE

Thought bubble HYPOMAGNESEMIA, magnesium leaking

WHAT IT ENCODES

Renal magnesium wasting can accompany the potassium loss. Hypomagnesemia perpetuates refractory hypokalemia (you cannot fully correct K until Mg is repleted) and contributes to the arrhythmia risk.

11. Symptoms box

WHAT YOU SEE

SYMPTOMS list: muscle weakness, polyuria, arrhythmia

WHAT IT ENCODES

Symptoms stem from hypokalemia: muscle weakness/cramps, fatigue, polyuria and polydipsia (hypokalemia causes nephrogenic diabetes insipidus), paresthesias, and arrhythmias (with ECG changes such as U waves, flattened T waves). Many patients are asymptomatic and present only with hard-to-control HTN.

12. End-organ damage

WHAT YOU SEE

Damaged heart, vessel, kidney, bone at the bottom

WHAT IT ENCODES

Aldosterone exerts BP-INDEPENDENT toxicity: myocardial and perivascular fibrosis (LVH, atrial fibrillation, heart failure), vascular remodeling/stiffness, glomerular hyperfiltration and proteinuria, and increased bone turnover. Patients with primary aldo have higher rates of MI, stroke, AF and renal injury than BP-matched essential hypertensives — a key reason to identify and treat it specifically with MR blockade or surgery.

BOARD TRAP

Assuming controlling the BP alone is enough — the aldosterone excess itself must be blocked (MRA) or removed (adrenalectomy) to prevent fibrotic end-organ damage.

13. Clinical pearl

WHAT YOU SEE

Yellow banner at the bottom

WHAT IT ENCODES

One-line summary: aldosterone retains Na (via ENaC), wastes K (via ROMK) and dumps acid (H⁺-ATPase) → HTN + hypokalemia + metabolic alkalosis; renin is suppressed (high ARR); aldosterone escape keeps Na near-normal and prevents edema; treat the MR (spironolactone/eplerenone) or remove the source (adrenalectomy).